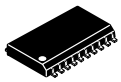
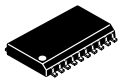
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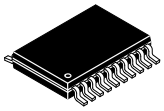
MM74HCT373/MM74HCT374



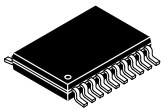
SOIC-20 WB
CASE 751D-05



SOIC-20, 300 mils
CASE 751BJ



TSSOP-20, 4.4x6.5
CASE 948AQ



TSSOP-20 WB
CASE 948E

General Description

The MM74HCT373 octal D-type latches and MM74HCT374 Octal D-type flip flops advanced silicon−gate CMOS technology, which provides the inherent benefits of low power consumption and wide power supply range, but are LS-TTL input and output characteristic & pin-out compatible. The 3-STATE outputs are capable of driving 15 LSTTL loads. All inputs are protected from damage due to static discharge by internal diodes to VCC and ground.

When the MM74HCT373 LATCH ENABLE input is HIGH, the Q outputs will follow the D inputs. When the LATCH ENABLE goes LOW, data at the D inputs will be retained at the outputs until LATCH ENABLE returns HIGH again. When a high logic level is applied to the OUTPUT CONTROL input, all outputs go to a high impedance state, regardless of what signals are present at the other inputs and the state of the storage elements.

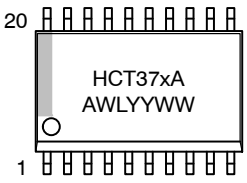
The MM74HCT374 are positive edge triggered flip-flops. Data at the D inputs, meeting the setup and hold time requirements, are transferred to the Q outputs on positive going transitions of the CLOCK (CK) input. When a high logic level is applied to the OUTPUT CONTROL (OC) input, all outputs go to a high impedance state, regardless of what signals are present at the other inputs and the state of the storage elements.

MM74HCT devices are intended to interface between TTL and NMOS components and standard CMOS devices. These parts are also plug in replacements for LS-TTL devices and can be used to reduce power consumption in existing designs.

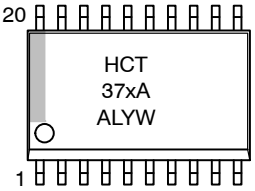
Features

- TTL Input Characteristic Compatible
- Typical Propagation Delay: 20 ns
- Low Input Current: 1 μA Maximum
- Low Quiescent Current: 160 μA Maximum
- Compatible with Bus−oriented Systems
- Output Drive Capability: 15 LS-TTL Loads
- These are Pb−Free Devices

MARKING DIAGRAMS



SOIC−20



TSSOP−20 WB

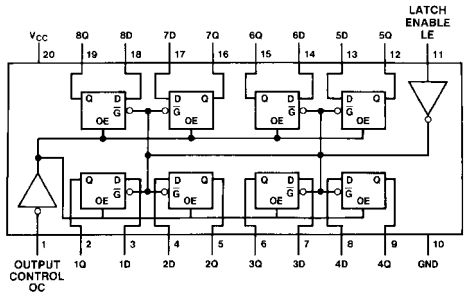
HCT37xA	= Specific Device Code x = 3 or 4
A	= Assembly Location
WL, L	= Wafer Lot
YY, Y	= Year
WW, W	= Work Week

ORDERING INFORMATION

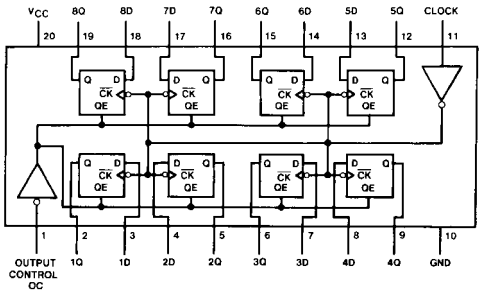
See detailed ordering and shipping information on page 7 of this data sheet.

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Connection Diagrams



MM74HCT373
(Top View)



MM74HCT374
(Top View)

Figure 1. Pin Assignments for SOIC and TSSOP

Truth Tables

MM74HCT373

Output Control	LE	Data	373 Output
L	H	H	H
L	H	L	L
L	L	X	Q_0
H	X	X	Z

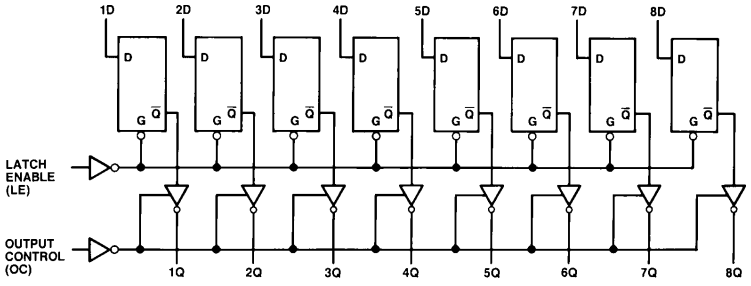
NOTES: H = HIGH Level
L = LOW Level
Q_0 = Level of output before steady−state input conditions were established.
Z = High Impedance

MM74HCT374

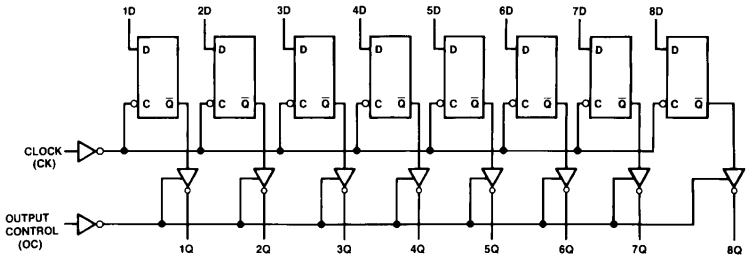
Output Control	Clock	Data	374 Output
L	↑	H	H
L	↑	L	L
L	L	X	Q_0
H	X	X	Z

NOTES: H = HIGH Level
L = LOW Level
X = Don't Care
↑ = Transition from LOW−to−HIGH
Z = High Impedance State
Q_0 = The level of the output before steady state input conditions were established.

Logic Diagrams



MM74HCT373



MM74HCT374

Figure 2. Logic Diagrams

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ABSOLUTE MAXIMUM RATINGS (Note 1)

Symbol	Parameter		Rating
V_{CC}	Supply Voltage		–0.5 to +6.5 V
V_{IN}	DC Input Voltage		–0.5 to V_{CC} + 0.5 V
V_{OUT}	DC Output Voltage		–0.5 to V_{CC} + 0.5 V
I_{IK}, I_{OK}	Clamp Diode Current		±20 mA
I_{OUT}	DC Output Current, per Pin		±35 mA
I_{CC}	DC V_{CC} or GND Current, per Pin		±70 mA
T_{STG}	Storage Temperature Range		–65 °C to +150 °C
P_D	Power Dissipation	S.O. Package only	500 mW
T_L	Lead Temperature (Soldering 10 Seconds)		260 °C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. Unless otherwise specified all voltages are referenced to ground.

RECOMMENDED OPERATING CONDITIONS

Symbol	Parameter	Min	Max	Unit
V_{CC}	Supply Voltage	4.5	5.5	V
V_{IN}, V_{OUT}	DC Input or Output Voltage	0	V_{CC}	V
T_A	Operating Temperature Range	–55	+125	°C
t_r, t_f	Input Rise or Fall Times		500	ns

Functional operation above the stresses listed in the Recommended Operating Ranges is not implied. Extended exposure to stresses beyond the Recommended Operating Ranges limits may affect device reliability.

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DC ELECTRICAL CHARACTERISTICS (V_{CC} = 5 V ± 10%, unless otherwise specified)

Symbol	Parameter	Conditions	T_A = 25 °C		T_A = −40 °C to 85 °C	T_A = −55 °C to 125 °C	Unit
			Typ	Guaranteed Limits			
V_{IH}	Minimum HIGH Level Input Voltage		–	2.0	2.0	2.0	V
V_{IL}	Maximum LOW Level Input Voltage		–	0.8	0.8	0.8	V
V_{OH}	Minimum HIGH Level Output Voltage	V_{IN} = V_{IH} or V_{IL} $\|I_{OUT}\|$ = 20 μA	V_{CC}	V_{CC} − 0.1	V_{CC} − 0.1	V_{CC} − 0.1	V
		V_{IN} = V_{IH} or V_{IL} $\|I_{OUT}\|$ = 6.0 mA, V_{CC} = 4.5 V	4.2	3.98	3.84	3.7	V
		V_{IN} = V_{IH} or V_{IL} $\|I_{OUT}\|$ = 7.2 mA, V_{CC} = 5.5 V	5.7	4.98	4.84	4.7	V
V_{OL}	Maximum LOW Level Voltage	V_{IN} = V_{IH} or V_{IL} $\|I_{OUT}\|$ = 20 μA	0	0.1	0.1	0.1	V
		V_{IN} = V_{IH} or V_{IL} $\|I_{OUT}\|$ = 6.0 mA, V_{CC} = 4.5 V	0.2	0.26	0.33	0.4	V
		V_{IN} = V_{IH} or V_{IL} $\|I_{OUT}\|$ = 7.2 mA, V_{CC} = 5.5 V	0.2	0.26	0.33	0.4	V
I_{IN}	Maximum Input Current	V_{IN} = V_{CC} or GND, V_{IH} or V_{IL}	–	±0.1	±1.0	±1.0	μA
I_{OZ}	Maximum 3−STATE Output Leakage Current	V_{OUT} = V_{CC} or GND, Enable = V_{IH} or V_{IL}	–	±0.5	±5.0	±10	μA
I_{CC}	Maximum Quiescent Supply Current	V_{IN} = V_{CC} or GND, I_{OUT} = 0 μA	–	8.0	80	160	μA
		V_{IN} = 2.4 V or 0.5 V (Note 2)	–	1.0	1.3	1.5	mA

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

2. Measured per pin. All others tied to V_{CC} or ground.

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AC ELECTRICAL CHARACTERISTICS

(V_{CC} = 5.0 V, T_A = 25 °C, $t_r = t_f$ = 6 ns, unless otherwise specified)

Symbol	Parameter	Conditions	Typ	Guaranteed Limit	Unit
t_{PHL}, t_{PLH}	Maximum Propagation Delay Data to Output	C_L = 45 pF	18	25	ns
t_{PHL}, t_{PLH}	Maximum Propagation Delay Latch Enable to Output	C_L = 45 pF	21	30	ns
t_{PZH}, t_{PZL}	Maximum Enable Propagation Delay Control to Output	C_L = 45 pF R_L = 1 kΩ	20	28	ns
t_{PHZ}, t_{PLZ}	Maximum Disable Propagation Delay Control to Output	C_L = 5 pF R_L = 1 kΩ	18	25	ns
t_W	Minimum Clock Pulse Width		–	16	ns
t_S	Minimum Setup Time Data to Clock		–	5	ns
t_H	Minimum Hold Time Clock to Data		–	10	ns

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

AC ELECTRICAL CHARACTERISTICS

(V_{CC} = 5.0 V ± 10%, $t_r = t_f$ = 6 ns, unless otherwise specified)

Symbol	Parameter	Conditions	T_A = 25 °C	T_A = −40 °C to 85 °C	T_A = −55 °C to 125 °C	Unit
			Typ	Guaranteed Limits		
t_{PHL}, t_{PLH}	Maximum Propagation Delay Data to Output	C_L = 50 pF	22	30	37	45	ns
		C_L = 150 pF	30	40	50	60	ns
t_{PHL}, t_{PLH}	Maximum Propagation Delay Latch Enable to Output	C_L = 50 pF	25	35	44	53	ns
		C_L = 150 pF	32	45	56	68	ns
t_{PZH}, t_{PZL}	Maximum Enable Propagation Delay Control to Output	C_L = 50 pF R_L = 1 kΩ	21	30	37	45	ns
		C_L = 150 pF R_L = 1 kΩ	30	40	50	60	ns
t_{PHZ}, t_{PLZ}	Maximum Disable Propagation Delay Control to Output	C_L = 50 pF R_L = 1 kΩ	21	30	37	45	ns
t_{THL}, t_{TLH}	Maximum Output Rise and Fall Time	C_L = 50 pF	8	12	15	18	ns
t_W	Minimum Clock Pulse Width		–	16	20	24	ns
t_S	Minimum Setup Time Data to Clock		–	5	6	8	ns
t_H	Minimum Hold Time Clock to Data		–	10	13	20	ns
C_{IN}	Maximum Input Capacitance		–	10	10	10	pF
C_{OUT}	Maximum Output Capacitance		–	20	20	20	pF
C_{PD}	Power Dissipation Capacitance (Note 3)	OC = V_{CC}	–	5	–	–	pF
		OC = GND	–	52	–	–	pF

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

3. C_{PD} determines the no load dynamic power consumption, $P_D = C_{PD} V_{CC}^2 f + I_{CC} V_{CC}$, and the no load dynamic current consumption, $I_S = C_{PD} V_{CC}^2 f + I_{CC}$.

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AC ELECTRICAL CHARACTERISTICS

(V_{CC} = 5.0 V, T_A = 25 °C, $t_r = t_f$ = 6 ns, unless otherwise specified)

Symbol	Parameter	Conditions	Typ	Guaranteed Limit	Unit
f_{MAX}	Maximum Clock Frequency		50	30	MHz
t_{PHL}, t_{PLH}	Maximum Propagation Delay Data to Output	C_L = 45 pF	20	32	ns
t_{PZH}, t_{PZL}	Maximum Enable Propagation Delay Control to Output	C_L = 45 pF R_L = 1 kΩ	19	28	ns
t_{PHZ}, t_{PLZ}	Maximum Disable Propagation Delay Control to Output	C_L = 5 pF R_L = 1 kΩ	17	25	ns
t_W	Minimum Clock Pulse Width		–	20	ns
t_S	Minimum Setup Time Data to Clock		–	5	ns
t_H	Minimum Hold Time Clock to Data		–	16	ns

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

AC ELECTRICAL CHARACTERISTICS

(V_{CC} = 5.0 V ± 10%, $t_r = t_f$ = 6 ns, unless otherwise specified)

Symbol	Parameter	Conditions	T_A = 25 °C	T_A = –40 °C to 85 °C	T_A = –55 °C to 125 °C	Unit
			Typ	Guaranteed Limits		
f_{MAX}	Minimum Clock Pulse Width		–	30	24	20	MHz
t_{PHL}, t_{PLH}	Maximum Propagation Delay to Output	C_L = 50 pF	22	36	45	48	ns
		C_L = 150 pF	30	46	57	69	ns
t_{PZH}, t_{PZL}	Maximum Enable Propagation Delay Control to Output	C_L = 50 pF R_L = 1 kΩ	21	30	37	45	ns
		C_L = 150 pF R_L = 1 kΩ	30	40	50	60	ns
t_{PHZ}, t_{PLZ}	Maximum Disable Propagation Delay Control to Output	C_L = 50 pF R_L = 1 kΩ	21	30	37	45	ns
t_{THL}, t_{TLH}	Maximum Output Rise and Fall Time	C_L = 50 pF	8	12	15	18	ns
t_W	Minimum Clock Pulse Width		–	16	20	24	ns
t_S	Minimum Setup Time Data to Clock		–	20	25	30	ns
t_H	Minimum Hold Time Clock to Data		–	5	5	5	ns
C_{IN}	Maximum Input Capacitance		–	10	10	10	pF
C_{OUT}	Maximum Output Capacitance		–	20	20	20	pF
C_{PD}	Power Dissipation Capacitance (Note 4)	OC = V_{CC}	–	5	–	–	pF
		OC = GND	–	58	–	–	pF

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

4. C_{PD} determines the no load dynamic power consumption, $P_D = C_{PD} V_{CC}^2 f + I_{CC} V_{CC}$, and the no load dynamic current consumption, $I_S = C_{PD} V_{CC}^2 f + I_{CC}$.

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ORDERING INFORMATION

Part Number	Package	Shipping†
MM74HCT373WMX	SOIC−20, Case 751BJ (Pb−Free and Halide−Free)	1000 Units / Tape & Reel
MM74HCT373MTCX	TSSOP−20 WB, Case 948E (Pb−Free)	2500 Units / Tape & Reel
MM74HCT374WM	SOIC−20 WB, Case 751D−05 (Pb−Free and Halide−Free)	38 Units / Tube
MM74HCT374WMX		1000 Units / Tape & Reel
MM74HCT374MTCX	TSSOP20, Case 948AQ−01 (Pb−Free)	2500 Units / Tape & Reel

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

REVISION HISTORY

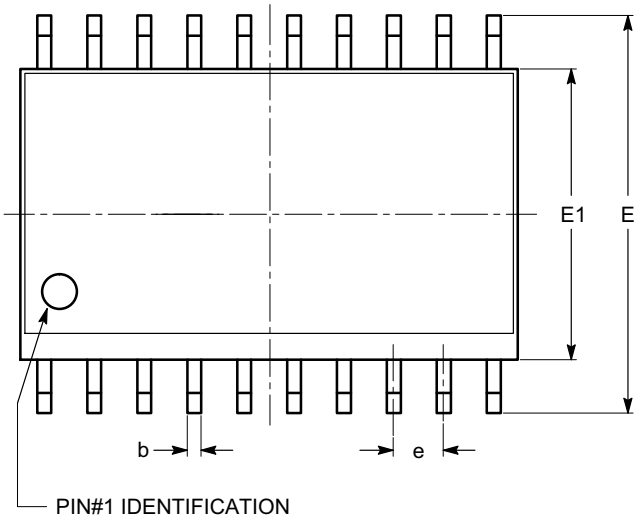
Revision	Description of Changes	Date
3	Changed Supply Voltage max range value from 7.0 V to 6.5 V.	2/20/2026

* Please note that this document has been previously updated prior to the inclusion of this revision history table and that the changes tracked only reflect what has occurred on the noted approval dates.

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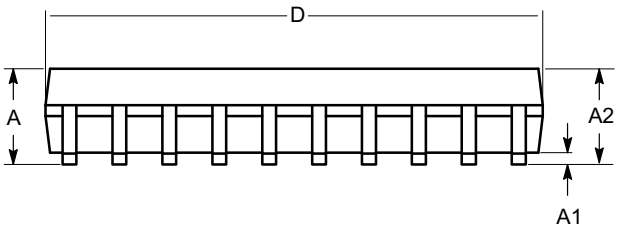
SOIC−20, 300 mils
CASE 751BJ
ISSUE O

DATE 19 DEC 2008

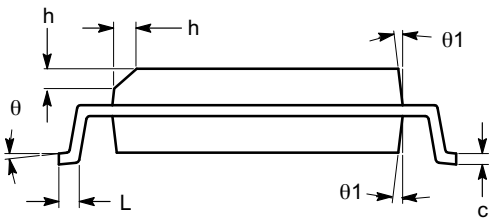


TOP VIEW

SYMBOL	MIN	NOM	MAX
A	2.36	2.49	2.64
A1	0.10		0.30
A2	2.05		2.55
b	0.31	0.41	0.51
c	0.20	0.27	0.33
D	12.60	12.80	13.00
E	10.01	10.30	10.64
E1	7.40	7.50	7.60
e		1.27 BSC	
h	0.25		0.75
L	0.40	0.81	1.27
θ	0°		8°
θ1	5°		15°



SIDE VIEW



END VIEW

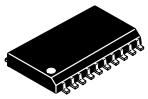
Notes:
(1) All dimensions are in millimeters. Angles in degrees.
(2) Complies with JEDEC MS-013.

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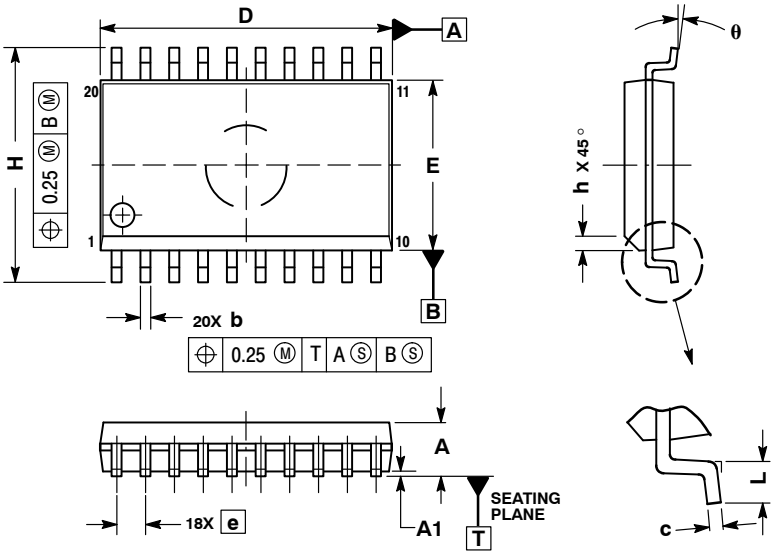
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SOIC−20 WB
CASE 751D−05
ISSUE H

DATE 22 APR 2015



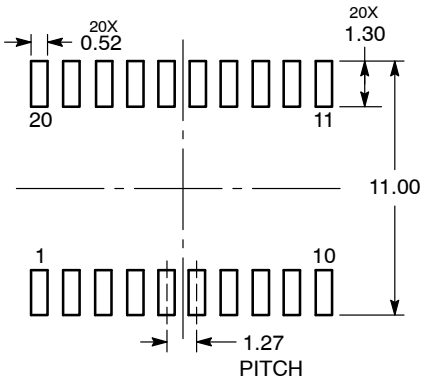
SCALE 1:1



NOTES:
1. DIMENSIONS ARE IN MILLIMETERS.
2. INTERPRET DIMENSIONS AND TOLERANCES PER ASME Y14.5M, 1994.
3. DIMENSIONS D AND E DO NOT INCLUDE MOLD PROTRUSION.
4. MAXIMUM MOLD PROTRUSION 0.15 PER SIDE.
5. DIMENSION B DOES NOT INCLUDE DAMBAR PROTRUSION. ALLOWABLE PROTRUSION SHALL BE 0.13 TOTAL IN EXCESS OF B DIMENSION AT MAXIMUM MATERIAL CONDITION.

	MILLIMETERS	
DIM	MIN	MAX
A	2.35	2.65
A1	0.10	0.25
b	0.35	0.49
c	0.23	0.32
D	12.65	12.95
E	7.40	7.60
e	1.27 BSC	
H	10.05	10.55
h	0.25	0.75
L	0.50	0.90
θ	0°	7°

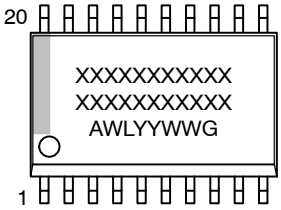
RECOMMENDED SOLDERING FOOTPRINT*



DIMENSIONS: MILLIMETERS

*For additional information on our Pb−Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

GENERIC MARKING DIAGRAM*



XXXXX = Specific Device Code
A = Assembly Location
WL = Wafer Lot
YY = Year
WW = Work Week
G = Pb−Free Package

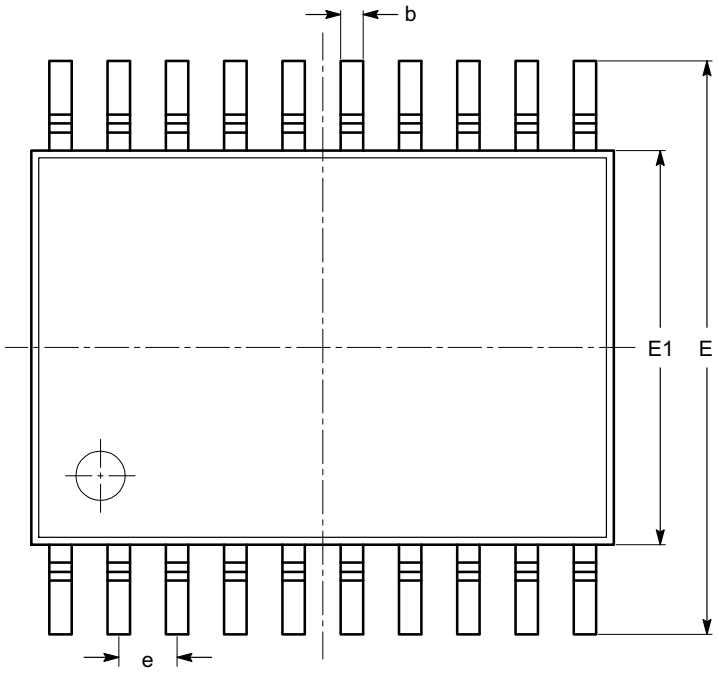
*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

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DESCRIPTION:	**SOIC−20 WB**		**PAGE 1 OF 1**

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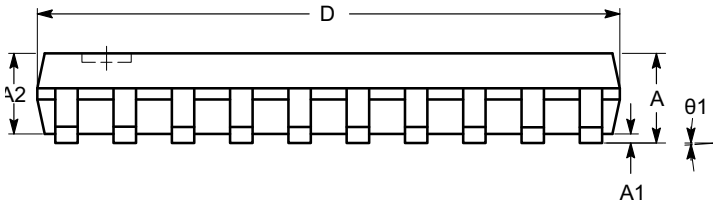
TSSOP20, 4.4x6.5
CASE 948AQ
ISSUE A

DATE 19 MAR 2009

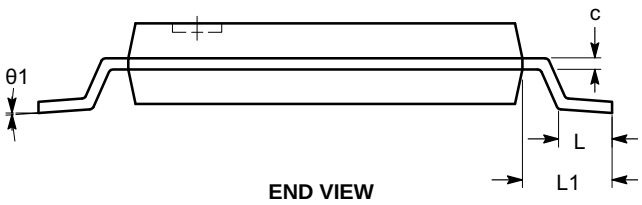


TOP VIEW

SYMBOL	MIN	NOM	MAX
A			1.20
A1	0.05		0.15
A2	0.80		1.05
b	0.19		0.30
c	0.09		0.20
D	6.40	6.50	6.60
E	6.30	6.40	6.50
E1	4.30	4.40	4.50
e		0.65 BSC	
L	0.45	0.60	0.75
L1		1.00 REF	
θ	0°		8°



SIDE VIEW



END VIEW

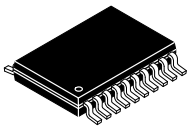
Notes:
(1) All dimensions are in millimeters. Angles in degrees.
(2) Complies with JEDEC MO-153.

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DESCRIPTION:	TSSOP20, 4.4X6.5	PAGE 1 OF 1

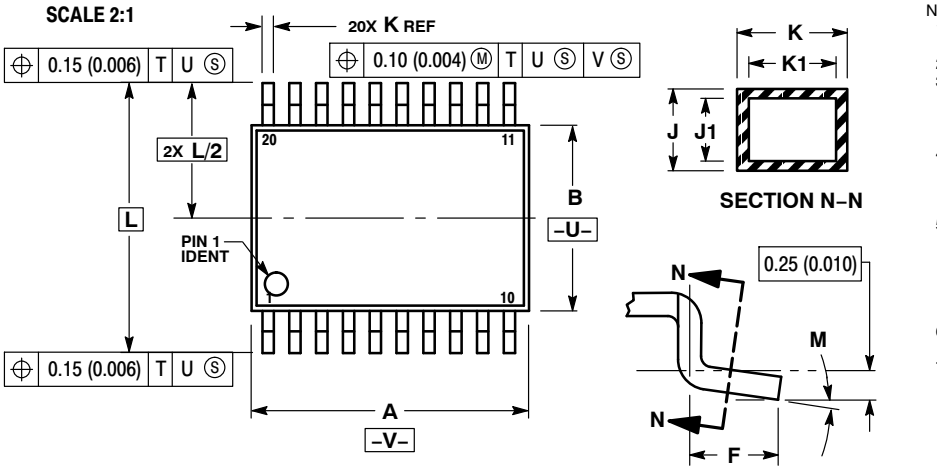
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TSSOP–20 WB
CASE 948E
ISSUE D

DATE 17 FEB 2016

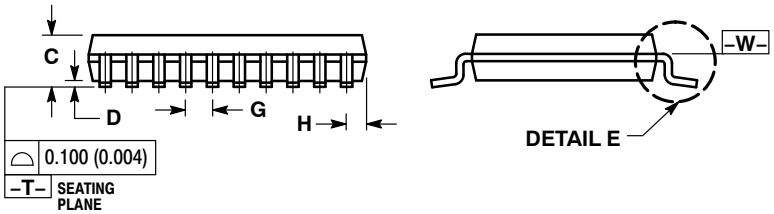


SCALE 2:1



SECTION N–N

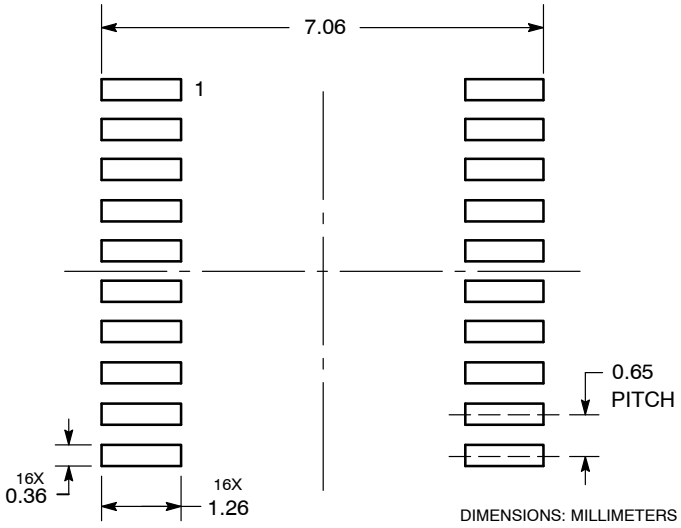
DETAIL E



NOTES:
1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: MILLIMETER.
3. DIMENSION A DOES NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BURRS. MOLD FLASH OR GATE BURRS SHALL NOT EXCEED 0.15 (0.006) PER SIDE.
4. DIMENSION B DOES NOT INCLUDE INTERLEAD FLASH OR PROTRUSION. INTERLEAD FLASH OR PROTRUSION SHALL NOT EXCEED 0.25 (0.010) PER SIDE.
5. DIMENSION K DOES NOT INCLUDE DAMBAR PROTRUSION. ALLOWABLE DAMBAR PROTRUSION SHALL BE 0.08 (0.003) TOTAL IN EXCESS OF THE K DIMENSION AT MAXIMUM MATERIAL CONDITION.
6. TERMINAL NUMBERS ARE SHOWN FOR REFERENCE ONLY.
7. DIMENSION A AND B ARE TO BE DETERMINED AT DATUM PLANE –W–.

	MILLIMETERS		INCHES	
DIM	MIN	MAX	MIN	MAX
A	6.40	6.60	0.252	0.260
B	4.30	4.50	0.169	0.177
C	---	1.20	---	0.047
D	0.05	0.15	0.002	0.006
F	0.50	0.75	0.020	0.030
G	0.65 BSC		0.026 BSC	
H	0.27	0.37	0.011	0.015
J	0.09	0.20	0.004	0.008
J1	0.09	0.16	0.004	0.006
K	0.19	0.30	0.007	0.012
K1	0.19	0.25	0.007	0.010
L	6.40 BSC		0.252 BSC	
M	0°	8°	0°	8°

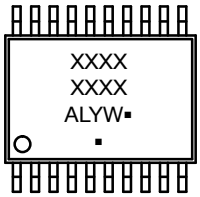
RECOMMENDED SOLDERING FOOTPRINT*



DIMENSIONS: MILLIMETERS

*For additional information on our Pb–Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

GENERIC MARKING DIAGRAM*



A = Assembly Location
L = Wafer Lot
Y = Year
W = Work Week
▪ = Pb–Free Package

(Note: Microdot may be in either location)

*This information is generic. Please refer to device data sheet for actual part marking. Pb–Free indicator, “G” or microdot “ ▪”, may or may not be present. Some products may not follow the Generic Marking.

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